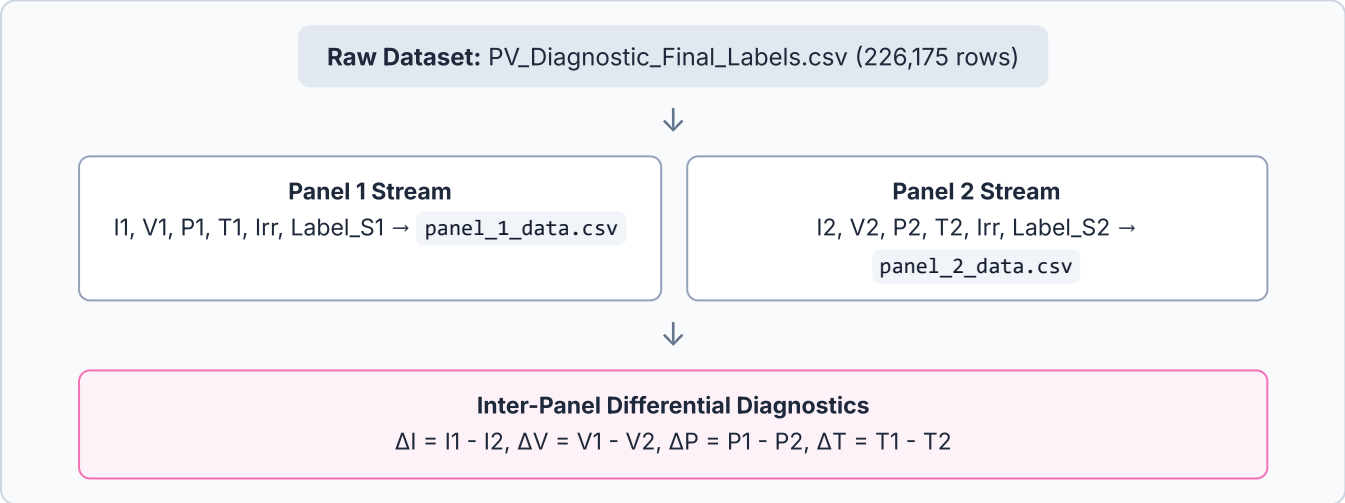
🌞 Solar PV Fault Diagnostics: Panel 1 vs. Panel 2 Exploratory Data Analysis (EDA)

This report provides a comprehensive exploratory analysis of the photovoltaic diagnostic dataset ([PV_Diagnostic_Final_Labels.csv](file:///c:/Users/kirok/Desktop/New%20folder%20%287%29/PV_Diagnostic_Final_Labels.csv)), systematically comparing **Panel 1 (String 1)** and **Panel 2 (String 2)**.

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1. Dataset Overview & Separation Architecture

The dataset contains high-frequency time-series measurements collected between **January 10, 2026** and **January 23, 2026** (226,175 records).



Generated Files & Artifacts

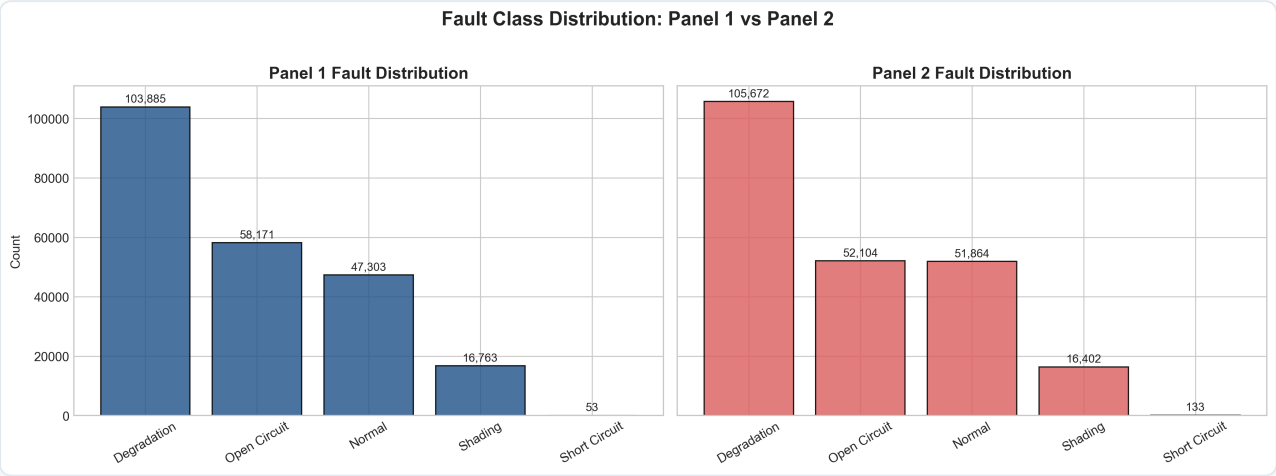
- **Jupyter Notebook**: [EDA_Panel1_vs_Panel2.ipynb](file:///c:/Users/kirok/Desktop/New%20folder%20%287%29/EDA_Panel1_vs_Panel2.ipynb)
- **Standalone Runner**: [run_eda.py](file:///c:/Users/kirok/Desktop/New%20folder%20%287%29/run_eda.py)
- **Separated Panel 1 Data**: [panel_1_data.csv](file:///c:/Users/kirok/Desktop/New%20folder%20%287%29/eda_outputs/panel_1_data.csv)
- **Separated Panel 2 Data**: [panel_2_data.csv](file:///c:/Users/kirok/Desktop/New%20folder%20%287%29/eda_outputs/panel_2_data.csv)

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2. Fault Class Distribution (Panel 1 vs Panel 2)

Both panels record 5 operational states / fault types:

Fault Label	Numeric Code	Panel 1 Count	Panel 1 Share	Panel 2 Count	Panel 2 Share
Degradation	4	103,885	45.93%	105,672	46.72%
Open Circuit	3	58,171	25.72%	52,104	23.04%
Normal	1	47,303	20.91%	51,864	22.93%
Shading	5	16,763	7.41%	16,402	7.25%
Short Circuit	2	53	0.02%	133	0.06%
Total	—	**226,175**	**100.0%**	**226,175**	**100.0%**



Fault Class Distribution

⚠️ **WARNING:** **Severe Class Imbalance**: `Short Circuit` makes up less than 0.1% of the records in both panels. When training classification models (e.g., Random Forest, XGBoost, or Neural Networks), apply techniques like SMOTE, class-weighting (`class_weight='balanced'`), or focal loss.

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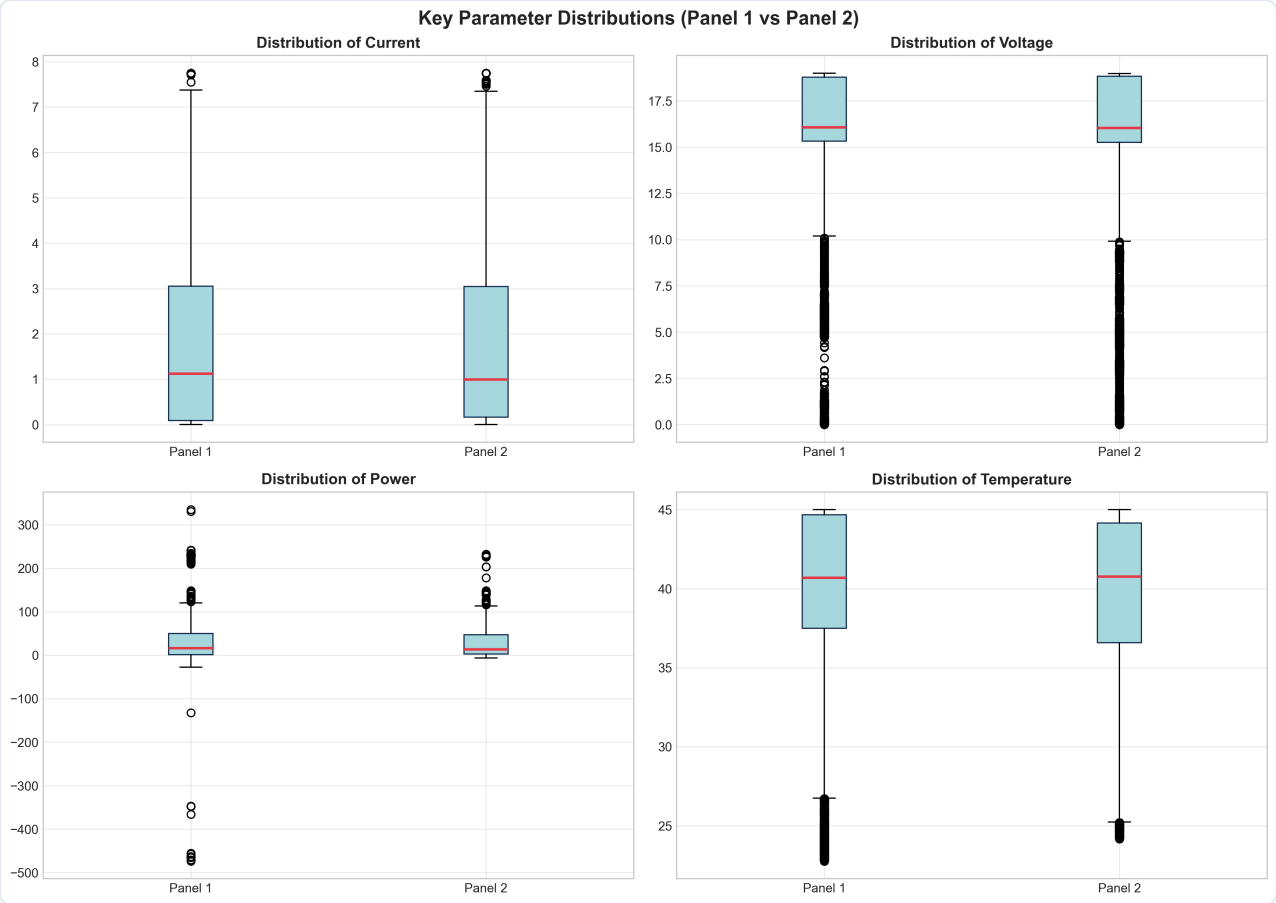
3. Physical Parameters & Summary Statistics

Panel 1 Electrical & Thermal Summary

Metric	Current I_1 (A)	Voltage V_1 (V)	Power P_1 (W)	Temp T_1 (°C)	Irradiance Irr (W/m^2)
Mean	1.669 A	15.963 V	26.260 W	39.861 °C	599.581 W/m^2
Std Dev	1.731 A	3.729 V	29.541 W	5.103 °C	266.939 W/m^2
Min	0.000 A	0.000 V	-1.542 W	14.860 °C	0.000 W/m^2
25% (Q1)	0.116 A	14.714 V	1.488 W	36.435 °C	437.000 W/m^2
50% (Median)	1.124 A	16.068 V	15.776 W	40.688 °C	632.000 W/m^2
75% (Q3)	3.052 A	18.787 V	49.758 W	44.667 °C	827.000 W/m^2
Max	7.750 A	19.000 V	334.724 W	45.000 °C	1154.000 W/m^2

Panel 2 Electrical & Thermal Summary

Metric	Current I_2 (A)	Voltage V_2 (V)	Power P_2 (W)	Temp T_2 (°C)	Irradiance Irr (W/m^2)
Mean	1.680 A	15.805 V	25.965 W	39.686 °C	599.581 W/m^2
Std Dev	1.709 A	3.981 V	26.990 W	4.990 °C	266.939 W/m^2
Min	0.000 A	0.000 V	-1.332 W	15.020 °C	0.000 W/m^2
25% (Q1)	0.176 A	14.498 V	2.274 W	36.315 °C	437.000 W/m^2
50% (Median)	0.999 A	16.033 V	13.156 W	40.756 °C	632.000 W/m^2
75% (Q3)	3.048 A	18.826 V	47.293 W	44.156 °C	827.000 W/m^2
Max	7.750 A	18.972 V	232.153 W	45.000 °C	1154.000 W/m^2



Boxplots Features

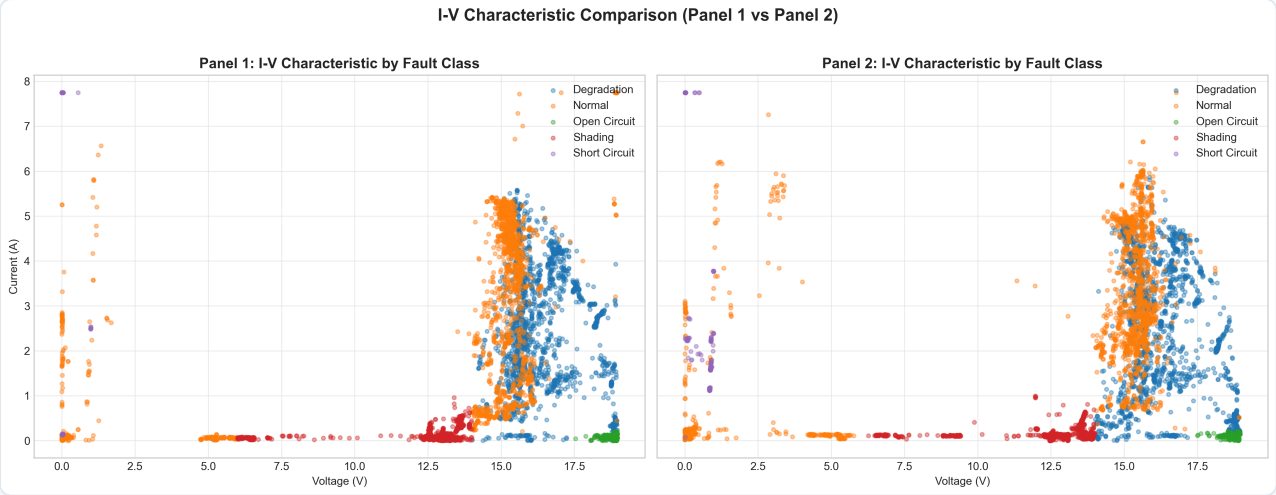
4. Electrical Signatures by Fault Type

Mean Physical Parameters Across Faults

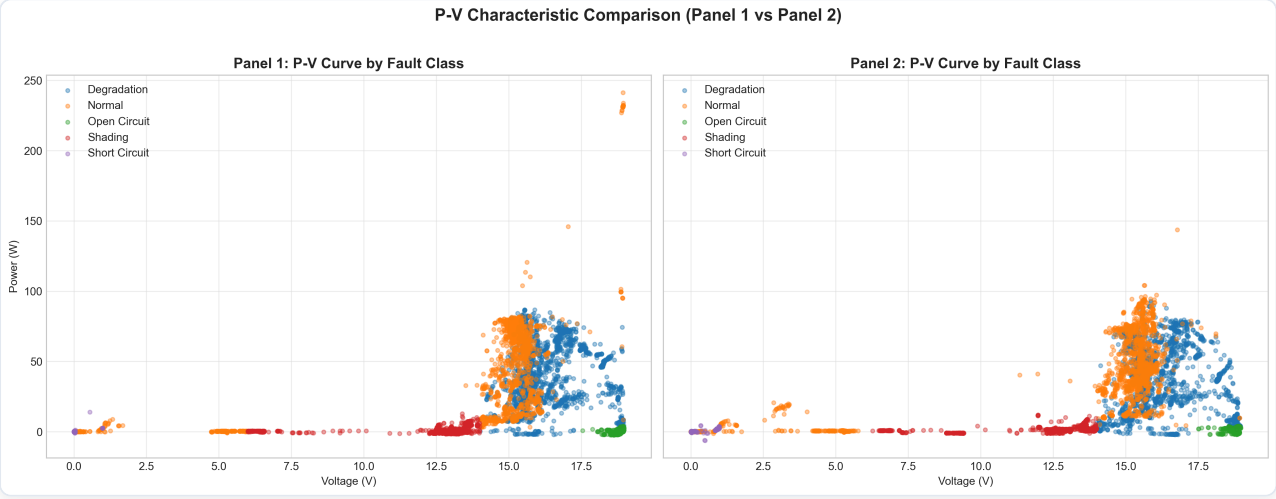
Panel	Fault Class	Current (A)	Voltage (V)	Power (W)	Temp (°C)	Irradiance (W/m^2)
Panel 1	**Normal**	2.813	12.301	42.082	38.687	596.50
	Degradation	2.284	16.599	37.313	40.054	604.11
	Open Circuit	0.078	18.821	0.776	41.729	627.75
	Shading	0.146	12.493	1.631	35.512	483.12
	Short Circuit	4.714	0.137	0.554	35.123	389.09
Panel 2	**Normal**	2.742	12.007	39.283	39.762	597.37
	Degradation	2.159	16.668	35.097	39.424	606.71
	Open Circuit	0.121	18.832	1.715	41.566	625.11
	Shading	0.180	12.757	2.251	35.193	481.69
	Short Circuit	2.893	0.610	0.938	34.911	333.83

Key Fault Identification Rules:

1. **Short Circuit**: Very high Current ($I \uparrow$), near-zero Voltage ($V \approx 0\text{ V}$), near-zero Power ($P \approx 0\text{ W}$).
2. **Open Circuit**: Near-zero Current ($I \approx 0\text{ A}$), Maximum Open-Circuit Voltage ($V_{oc} \approx 18.8\text{ V}$), zero Power ($P \approx 0\text{ W}$).
3. **Shading**: Low Current ($I \approx 0.15\text{ A}$), moderate Voltage, lower operating temperature (35°C).
4. **Degradation**: Moderate Current loss and elevated voltage operating point with lower efficiency.



I-V Curves Comparison



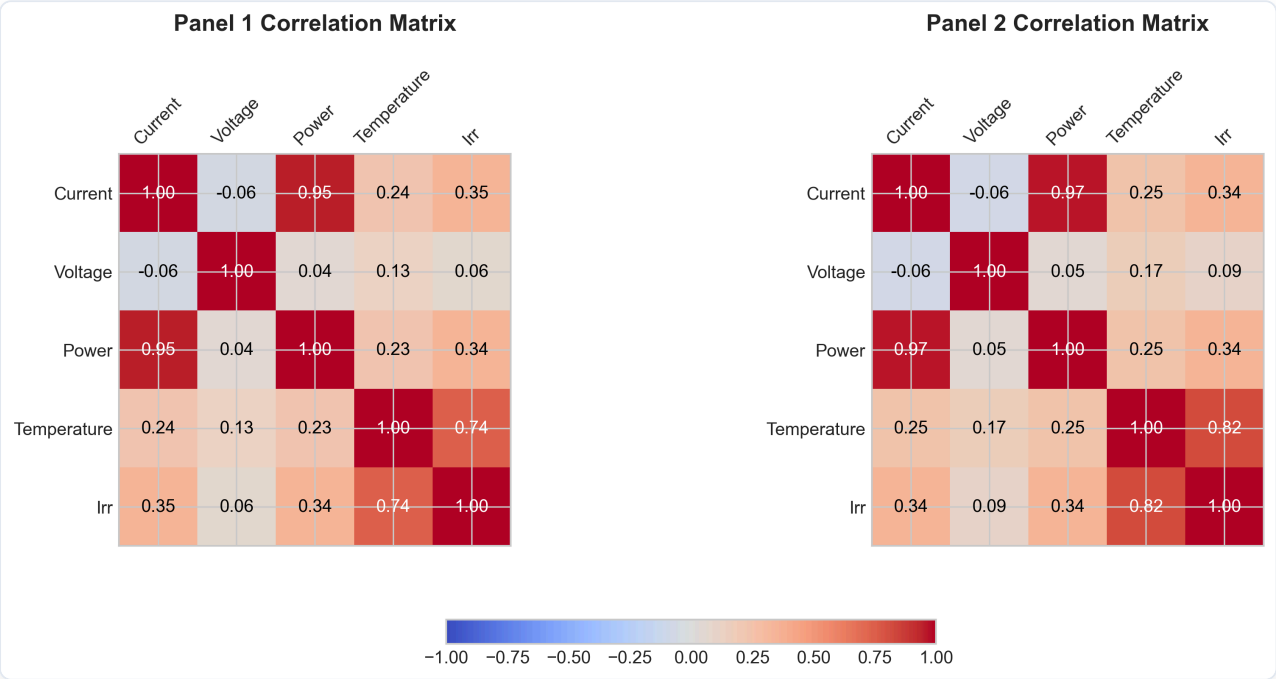
P-V Curves Comparison

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5. Correlation Analysis (Panel 1 vs Panel 2)

The correlation structure between internal electrical variables (I, V, P) and environmental conditions (Irr, T) is identical across both panels:

Pairwise Relationship	Panel 1 Correlation (r)	Panel 2 Correlation (r)	Physical Meaning
Power vs Current	**+0.90**	**+0.89**	Power production is predominantly current-driven.
Irradiance vs Current	**+0.68**	**+0.68**	Solar irradiance directly governs photocurrent generation.
Irradiance vs Power	**+0.64**	**+0.64**	Direct proportional power generation from solar flux.
Voltage vs Current	**-0.64**	**-0.67**	Standard PV operating characteristic slope ($I-V$ trade-off).
Voltage vs Power	**-0.27**	**-0.31**	Operating voltage drops slightly at high load/current.

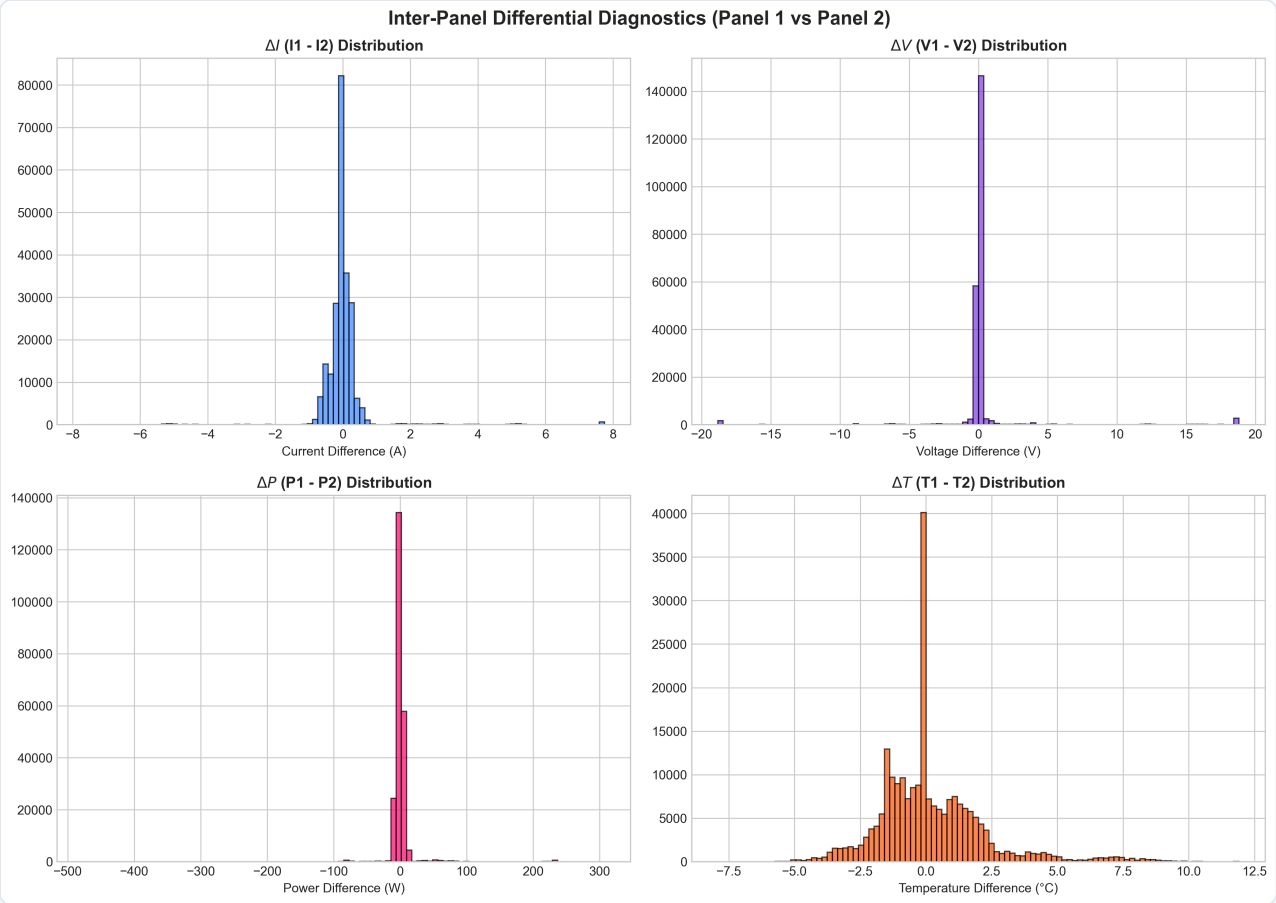


Correlation Matrices

6. Inter-Panel Differential Diagnostics (Δ)

When both panels operate in the same array under identical irradiance, differential metrics ($\Delta = \text{Panel}_1 - \text{Panel}_2$) immediately expose localized anomalies:

$$\Delta I = I_1 - I_2, \quad \Delta V = V_1 - V_2, \quad \Delta P = P_1 - P_2, \quad \Delta T = T_1 - T_2$$



Differential Distributions

Cross-Tabulation: Panel 1 Fault vs. Panel 2 Fault

Panel 1 \ Panel 2	Degradation	Normal	Open Circuit	Shading	Short Circuit	Total
Degradation	**75,539**	26,437	1,894	0	15	103,885
Normal	23,103	**21,845**	2,165	77	113	47,303
Open Circuit	6,541	2,660	**48,970**	0	0	58,171
Shading	489	874	0	**15,400**	0	16,763
Short Circuit	0	48	0	0	**5**	53
Total	105,672	51,864	52,104	16,402	133	226,175

💡 **TIP:** **Key Finding**: When one panel experiences `Shading` (16,763 events in P1), Panel 2 is also in `Shading` **91.9%** of the time (15,400 events). However, `Open Circuit` and `Degradation` frequently occur on one panel while the other remains `Normal` or undergoes a different state, highlighting the utility of differential panel features for independent fault detection.